

SIEMLab — Commands & Configuration Reference

This document contains every command and configuration file used during the lab build, organized by host and by phase. Use it alongside `01-SIEMLab-Architecture-and-Setup.md` for the reasoning behind each step.

A. VMware — Network Setup

Creating the custom Host-only network:

1. Edit → Virtual Network Editor → Add Network → select `VMnet2`
2. Type: **Host-only**
3. Uncheck "Connect a host virtual adapter"
4. Uncheck "Use local DHCP service"
5. Subnet: `192.168.56.0/24`

Static IP allocation table:

Host	Interface	IP	Gateway	DNS
DC	Host-only	192.168.56.10	—	192.168.56.10
DC	NAT	DHCP-assigned	DHCP-assigned	192.168.56.10 (manual)
Win10	Host-only	192.168.56.20	—	192.168.56.10
Win10	NAT	DHCP-assigned	DHCP-assigned	192.168.56.10 (manual)
Ubuntu	Host-only	192.168.56.30	—	192.168.56.10
Ubuntu	NAT	DHCP-assigned	DHCP-assigned	192.168.56.10 (no override)

B. Windows Server (DC) — PowerShell Commands

B.1. Rename the host & set the static IP (via GUI or PowerShell)

```
Rename-Computer -NewName "DC01" -Restart
```

B.2. Install the AD DS role and promote to Domain Controller

```
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
```

```
Install-ADDSForest `
  -DomainName "cslab.local" `
  -DomainNetbiosName "CSLAB" `
  -InstallDns:$true `
  -Force:$true
```

B.3. Configure the DNS forwarder (external resolution)

```
Add-DnsServerForwarder -IPAddress "8.8.8.8", "1.1.1.1"

# Verify
Get-DnsServerForwarder
```

Or via GUI: **Server Manager** → **Tools** → **DNS** → right-click server → **Properties** → **Forwarders** tab → **Edit** → add 8.8.8.8, 1.1.1.1

B.4. Disable Dynamic DNS Registration on the NAT interface

GUI: **Network Adapter Properties** (NAT interface) → **IPv4 Properties** → **Advanced** → **DNS** tab → uncheck "Register this connection's addresses in DNS"

B.5. Full network configuration check

```
ipconfig /all
```

B.6. Verify the routing table (confirm a single default route via NAT)

```
route print
```

B.7. Test DNS resolution + internet connectivity

```
# Resolve an external domain (via forwarder)
nslookup google.com

# Resolve an internal domain
nslookup cslab.local

# Test actual internet reachability (not just DNS)
Test-NetConnection google.com -Port 443

# Confirm reachability to self via the internal interface
Test-NetConnection 192.168.56.10 -Port 445
```

C. Windows 10 Client — PowerShell Commands

C.1. Set the static IP + DNS (via GUI or PowerShell)

```
New-NetIPAddress -InterfaceAlias "Ethernet0" -IPAddress "192.168.56.20" -  
PrefixLength 24  
Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses  
"192.168.56.10"
```

C.2. Join the domain

```
Add-Computer -DomainName "cslab.local" -Restart
```

C.3. Install Sysmon with the SwiftOnSecurity configuration

```
mkdir C:\Tools  
cd C:\Tools  
  
# Download Sysmon from Microsoft Sysinternals  
Invoke-WebRequest -Uri "https://download.sysinternals.com/files/Sysmon.zip"  
-OutFile "Sysmon.zip"  
Expand-Archive -Path "Sysmon.zip" -DestinationPath "Sysmon"  
  
# Download the SwiftOnSecurity configuration  
Invoke-WebRequest -Uri  
"https://raw.githubusercontent.com/SwiftOnSecurity/sysmon-  
config/master/sysmonconfig-export.xml" -OutFile "sysmonconfig.xml"  
  
# Install Sysmon with the configuration  
cd C:\Tools\Sysmon  
.\Sysmon64.exe -i C:\Tools\sysmonconfig.xml -accepteula  
  
# Verify the service  
Get-Service sysmon64
```

View logs at: Event Viewer → Applications and Services Logs → Microsoft → Windows
→ Sysmon → Operational

C.4. Install the Splunk Universal Forwarder

Download the .msi installer from: https://www.splunk.com/en_us/download/universal-forwarder.html

During installation (GUI installer):

- **Receiving Indexer:** 192.168.56.30:9997

- **Deployment Server:** 192.168.56.30:8089

C.5. Repointing the Deployment Server after install (without re-running the installer)

```
cd "C:\Program Files\SplunkUniversalForwarder\bin"  
.\splunk.exe set deploy-poll 192.168.56.30:8089  
.\splunk.exe restart
```

C.6. Verify the forwarder received the app from the Deployment Server

```
dir "C:\Program Files\SplunkUniversalForwarder\etc\apps\sysmon_inputs\local"
```

C.7. Check the forwarder log when troubleshooting connection issues

```
Get-Content "C:\Program  
Files\SplunkUniversalForwarder\var\log\splunk\splunkd.log" -Tail 50
```

C.8. Install Atomic Red Team (Invoke-AtomicRedTeam)

Set execution policy for the current user only — allows local/unsigned scripts to run without widening policy machine-wide:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Run as Administrator — installs the module and downloads the full atomics test case library:

```
IEX (IWR 'https://raw.githubusercontent.com/redcanaryco/invoke-  
atomicredteam/master/install-atomicredteam.ps1' -UseBasicParsing)  
Install-AtomicRedTeam -getAtomics -Force
```

- `-getAtomics` downloads all MITRE ATT&CK technique test cases to `C:\AtomicRedTeam\atomics`.
- `-Force` overwrites any prior install without prompting.

Auto-load the module on every new PowerShell session via `$PROFILE`:

```
notepad $PROFILE
```

Add this line and save:

```
Import-Module "$env:USERPROFILE\Documents\WindowsPowerShell\Modules\Invoke-AtomicRedTeam\Invoke-AtomicRedTeam.psd1" -Force
```

Reload the profile in the current session (new sessions pick it up automatically):

```
. $profile
```

Verify installation:

```
Get-Module Invoke-AtomicRedTeam
$PSDefaultParameterValues
Invoke-AtomicTest T1059.001 -ShowDetails
```

- `Get-Module` confirms the module is loaded (version, path, exported commands) — useful evidence for the lab report.
- `-ShowDetails` (not `-ShowDetailsBrief`) prints the full technique definition — command, arguments, dependencies — **without executing it**, confirming the atomics folder loaded and parsed correctly.

D. Ubuntu Server — Bash Commands

D.1. Check interfaces & set the static IP via Netplan (Host-only)

```
ip a
sudo nano /etc/netplan/00-installer-config.yaml
```

```
network:
  version: 2
  ethernets:
    ens33:                                # host-only interface name - adjust to
match your system                        match your system
    dhcp4: no
    addresses:
      - 192.168.56.30/24
    nameservers:
      addresses: [192.168.56.10]
```

```
sudo netplan apply
ip a
ping 192.168.56.10
```

D.2. Add the NAT interface (dual-homing) — update Netplan

```
network:
  version: 2
  ethernets:
    ens33:                                # host-only
      dhcp4: no
      addresses:
        - 192.168.56.30/24
      nameservers:
        addresses: [192.168.56.10]
    ens37:                                # NAT - interface name may vary by
system
  dhcp4: yes
  dhcp4-overrides:
    use-dns: false                        # keep DNS pointed at the DC; don't let
NAT override it
```

```
sudo netplan apply
```

D.3. Verify routing (only one default route, via NAT)

```
ip route
```

Expected output:

```
forsecservice@cybersecservice:~/cslab/splunk$ ip route
default via 192.168.146.2 dev ens37 proto dhcp src 192.168.146.157 metric 100
172.19.0.0/16 dev br-fbf4c598e3cb proto kernel scope link src 172.19.0.1
192.168.56.0/24 dev ens33 proto kernel scope link src 192.168.56.30
192.168.146.0/24 dev ens37 proto kernel scope link src 192.168.146.157 metric 100
192.168.146.2 dev ens37 proto dhcp scope link src 192.168.146.157 metric 100
```

D.4. Two-way connectivity test (internal + internet)

```
ping 192.168.56.10      # internal - should reach the DC
ping 8.8.8.8            # internet reachability
sudo apt update         # real-world functional test
cat /etc/resolv.conf    # confirm DNS still points to 192.168.56.10
nslookup google.com 192.168.56.10 # test the forwarder from the DC's side
```

D.5. System update + Docker install

```
sudo apt update && sudo apt upgrade -y

curl -fsSL https://get.docker.com -o get-docker.sh
sudo sh get-docker.sh
```

```
# Allow running Docker without sudo
sudo usermod -aG docker $USER
newgrp docker

# Verify
docker run hello-world
docker compose version
```

D.6. Docker Compose — Splunk

```
mkdir -p ~/cslab/splunk
cd ~/cslab/splunk
nano docker-compose.yaml
```

```
services:
  splunk:
    image: splunk/splunk:latest
    container_name: splunk
    hostname: splunk
    restart: unless-stopped
    ports:
      - "8000:8000"    # Web UI
      - "9997:9997"    # Receiver port for forwarders
      - "8089:8089"    # Management port / REST API
    environment:
      - SPLUNK_PASSWORD=YourStrongP@ssw0rd
      - SPLUNK_START_ARGS=--accept-license
      - SPLUNK_GENERAL_TERMS=--accept-sgt-current-at-splunk-com
    volumes:
      - splunk_etc:/opt/splunk/etc
      - splunk_var:/opt/splunk/var

volumes:
  splunk_etc:
  splunk_var:
```

```
docker compose up -d
docker compose logs -f splunk    # follow until "Ansible playbook complete"
appears
```

If changes are needed and the container must be recreated (data persists via the named volumes):

```
docker compose down
docker compose up -d
```

D.7. Docker Compose — Portainer (separate file, not merged with Splunk)

```
mkdir -p ~/cslab/portainer
cd ~/cslab/portainer
nano docker-compose.yml
```

```
services:
  portainer:
    image: portainer/portainer-ce:latest
    container_name: portainer
    restart: unless-stopped
    ports:
      - "9443:9443" # Web UI (HTTPS)
    volumes:
      - /var/run/docker.sock:/var/run/docker.sock
      - portainer-data:/data

volumes:
  portainer-data:
```

```
docker compose up -d
```

Access at: `https://192.168.56.30:9443` (accept the self-signed certificate warning).

D.8. Install Volatility 3 (Python virtual environment — prep for Week 6 memory forensics)

```
mkdir ~/tools/volatility3
git clone https://github.com/volatilityfoundation/volatility3.git
~/tools/volatility3
cd ~/tools/volatility3

python3 -m venv ~/tools/volatility-venv
source ~/tools/volatility-venv/bin/activate

pip install -e ".[dev]"
```

- Installed in its own venv (`~/tools/volatility-venv`), kept isolated from system Python and from any packages used by other tools on the host.
- `pip install -e ".[dev]"` installs Volatility 3 in editable mode with the `dev` extras.

Verify:


```
python3 vol.py -h
```

Reactivate the venv in future sessions:

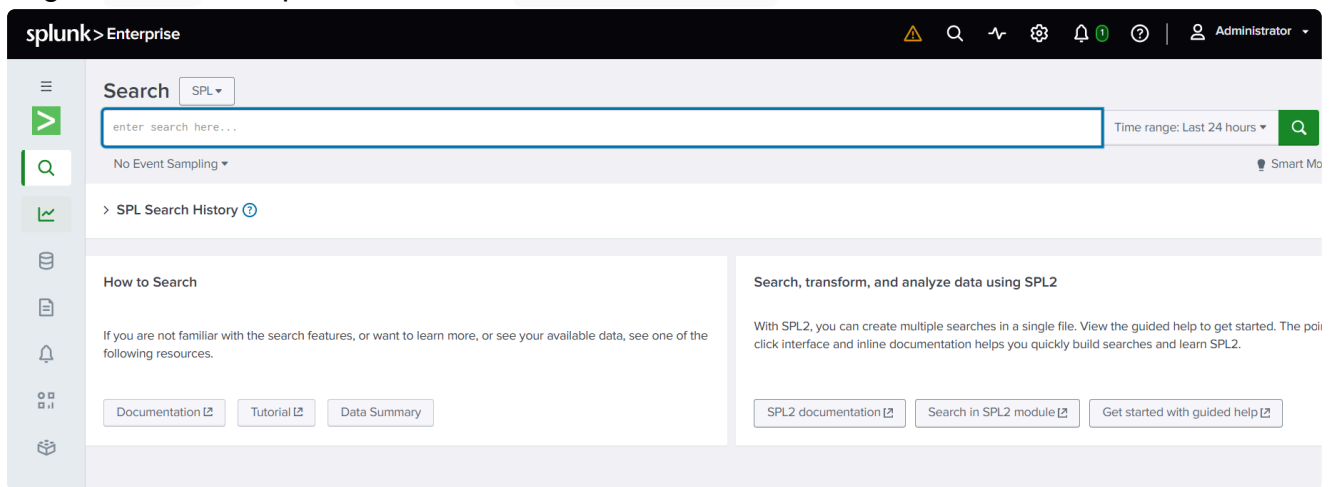
```
source ~/tools/volatility-venv/bin/activate
```

E. Splunk — Configuration via Web UI

E.1. Access the Splunk UI

Access at: `http://192.168.56.30:8000`

Login: `admin` / the password set in `SPLUNK_PASSWORD`.



E.2. Create the deployment app containing the Sysmon + Security log config

Enter the container:

```
docker exec -it splunk bash
```

Create the app structure:

```
mkdir -p /opt/splunk/etc/deployment-apps/sysmon_inputs/local
cd /opt/splunk/etc/deployment-apps/sysmon_inputs/local

cat > inputs.conf << 'EOF'
[WinEventLog://Microsoft-Windows-Sysmon/Operational]
disabled = 0
index = sysmon
sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

[WinEventLog://Security]
```

```
disabled = 0
index = windows_security
sourcetype = WinEventLog:Security
EOF

exit
```

E.3. Create the Server Class under Agent Management

Settings → Agent management → Server Classes tab → New Server Class

- Name: win10_clients
- Apps tab → Add Apps → select sysmon_inputs → Save
- Clients tab → Add Clients → Host-Victim (Win10 hostname) → Save

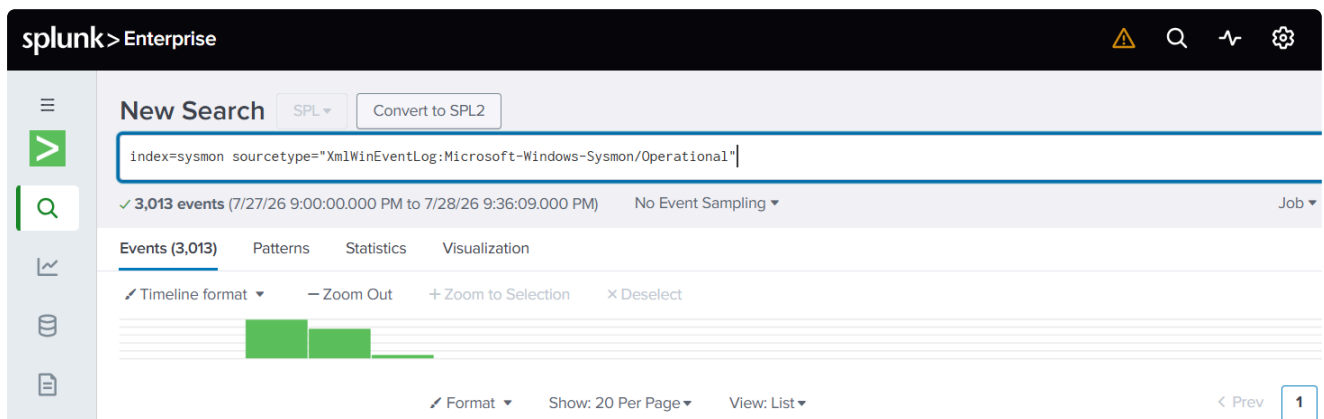
E.4. Verify client check-in

Settings → Agent management → Clients tab

Confirm the Windows 10 hostname appears, with the Apps column showing `sysmon_inputs` as deployed ("Up to date").

E.5. Search to verify logs are flowing in

```
index=sysmon sourcetype="XmlWinEventLog:Microsoft-Windows-
Sysmon/Operational"
```



F. Troubleshooting — Log Forwarding & Permissions

F.1. Lessons learned

- Learned how to configure log forwarding from the Windows 10 client to the Splunk server end-to-end.
- Learned a systematic troubleshooting approach for forwarder issues: starting from basic connectivity checks and working down to the forwarder's own internal log, `C:\Program Files\SplunkUniversalForwarder\var\log\splunk\splunkd.log`.

- Discovered a permissions issue specific to the Splunk Universal Forwarder: by default, its service account may lack read access to certain sensitive event log channels — notably the Sysmon operational log and the Windows Security event log — which silently prevents those sources from being forwarded even when the input is configured correctly.

F.2. Troubleshooting checklist

1. **Confirm network connectivity** from the client to the indexer/deployment server (port 9997 for data, 8089 for management):

```
Test-NetConnection 192.168.56.30 -Port 9997
Test-NetConnection 192.168.56.30 -Port 8089
```

2. **Confirm the deployment app was received** (see C.6 above).
3. **Check the forwarder's internal log** for errors (see C.7 above) — this is where read-permission failures on Sysmon/Security event channels surface.
4. **Check event log read permissions** if `splunkd.log` reports access-denied errors on `WinEventLog` inputs — the forwarder's service account needs membership in the local `Event Log Readers` group (or equivalent read rights) to access the Security and Sysmon operational channels.
5. **Verify data arrival in Splunk** via the search in E.5 once connectivity, deployment, and permissions are all confirmed.